3-D Digital Compute-in-Memory Benchmark With A5 CFET Technology: An Extension to Lookup-Table-Based Design

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***Abstract*—Digital compute-in-memory (DCIM) has emerged as a promising solution to address scalability and accuracy challenges in analog compute-in-memory (ACIM) for next-generation AI hardware acceleration. In this work, we present a comprehensive device-to-system codesign process for the two proposed 3-D DCIM architectures at the projected 5 angstrom (A5) complementary FET (CFET) technology node: 1) 3-D DCIM based on 8T DCIM bit cell and 2) lookup-table (LUT)-based 3-D DCIM. A novel A5 CFET-based 8T DCIM bit cell (6T SRAM +2T AND gate) is proposed to improve total footprint and latency over the conventional 10T DCIM bit cell, and its functionality is verified through technology computer-aided design (TCAD) simulation. For macro- and system-level evaluation of the proposed 3-D DCIM architectures, an extended NeuroSim V1.4 framework is developed, the first compute-in-memory (CIM) benchmark framework enabling CIM simulation at the A5 CFET technology node. We demonstrate that the proposed 3-D DCIM with 8T DCIM bit cell at the A5 CFET technology node can achieve 8.2× improvement in figure of merit (FOM) (=TOPS/W × TOPS/mm^2) over the state-of-the-art 3-nm FinFET-based DCIM design. The LUT-based 3-D DCIM design is additionally proposed to achieve further power consumption reduction from the 8T DCIM bit-cell-based 3-D DCIM. LUT-based 3-D DCIM achieves a 44% reduction in energy consumption compared to the conventional 10T DCIM bit-cell-based 3-D DCIM. Our findings suggest the significant implications for technology scaling below 1 nm in high-performance DCIM design.**

***Index Terms*—5 angstrom (A5) node, complementary FET (CFET), digital compute-in-memory (DCIM), NeuroSim, technology scaling.**

I. Introduction

A VARIETY of architectures for compute-in-memory (CIM) have been explored to efficiently handle intensive multiply-and-accumulate (MAC) operations for data-intensive workloads such as deep neural networks (DNNs) [1]. Until recently, the mainstream CIM architecture explored in the literature had been analog CIM (ACIM), which employs SRAM or nonvolatile memory devices to perform MAC operations with current or charge accumulation. Most ACIM designs, however, suffer from limited scalability toward leading-edge nodes. For example, state-of-the-art ACIM prototype chips have only been demonstrated down to 22 nm (for RRAM [2]) and 7 nm (for SRAM [3]). Meanwhile, the digital CIM (DCIM) paradigm [4] has recently been suggested at a 22-nm technology node to harness technology scaling and address the challenges of ACIM, such as signal integrity and accuracy degradation from process-voltage-temperature (PVT) variations. The key concept of DCIM is the introduction of a digital bit cell, typically consisting of a storage cell (6T SRAM) followed by a local multiplier [4], [5]. Since the initial demonstration, several DCIM prototype chips with different digital bit-cell designs have been developed by leading foundries such as TSMC. Recently, state-of-the-art DCIM chips have been developed at the 3-nm FinFET node [5]. However, the scalability of DCIM designs in the future technology nodes that employ stacked nanosheets and ultimately complementary FET (CFET) technology remains largely unexplored. This article is an extension of the conference paper [6], where we focus on CFET technology integration for DCIM and propose a novel CFET-based 8T DCIM bit cell implemented at 5 angstrom (A5) CFET technology node to reduce the area overhead of MAC units. The proposed A5 CFET-based 8T DCIM bit-cell design leverages the 3-D stackability of pMOS and nMOS, improving total footprint, latency, and energy consumption over previous technology nodes and conventional 10T DCIM bit-cell design. In this journal extension, we additionally propose lookup-table (LUT)-based MAC operation to further enhance dynamic power consumption and area efficiency from our prior work.

This article presents a device-to-system codesign process of two novel 3-D DCIM architectures using the extended NeuroSim V1.4 framework [7]. First, we propose and evaluate a 3-D DCIM architecture consisting of an A5 CFET-based 3-D 8T DCIM bit cell (6T SRAM +2T AND gate). As an extension of the first proposed architecture, we additionally present a 3-D DCIM architecture utilizing LUT-based MAC, eliminating local multipliers and the first stage of the adder tree at the A5 CFET technology node. The A5 CFET device model for the proposed 3-D 8T DCIM bit cell, as well as the peripheral circuits, is simulated and calibrated in Sentaurus technology computer-aided design (TCAD). The device design parameters are calibrated with relevant physical models to

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meet the International Roadmap for Devices and Systems (IRDS) performance standard at the A5 CFET technology node. The functionality of the calibrated 3-D 8T DCIM bit cell is validated through mixed-mode simulation in Sentaurus TCAD. For a macro- and system-level evaluation of the proposed 3-D DCIM architectures, we develop an extended NeuroSim V1.4 framework [7], which incorporates the calibrated device and circuit-level parameters for A5 CFET and 3-D 8T DCIM bit cell. All the peripheral circuits for the CIM inference operations, such as shift add and buffer, are scaled using the calibrated A5 CFET device parameters. To the best of our knowledge, this is the first benchmark framework that allows early stage macro- and system-level evaluation and design exploration of A5 CFET-based circuit applications. Using the extended NeuroSim V1.4, we conduct a macro- and system-level performance comparison between 3-D DCIMs based on: 1) conventional MAC with conventional 10T DCIM bit cell; 2) conventional MAC with the proposed 3-D 8T DCIM bit cell (the first proposed 3-D DCIM); and 3) LUT-based MAC without local multipliers (the second proposed 3-D DCIM) at A5 CFET technology node. By evaluating these architectures at macro- and system-level using the extended NeuroSim V1.4, we not only show the superior performance of the proposed 3-D DCIMs over the prior DCIM designs but also highlight significant implications for the adoption of A5 CFET technology node in DCIM design. The contributions of this study are summarized as follows.

1) At the device level, CFET at the A5 technology node is modeled and parametrically calibrated using Sentaurus TCAD.
2) A novel 3-D 8T DCIM bit cell with 6T SRAM +2T AND gate is proposed and verified through TCAD mixed-mode simulation. In a DCIM macro, the proposed bit cell at the A5 CFET technology node improves area density/latency/energy efficiency by ~60.9%, ~19%, and ~61.7%, respectively, compared to a conventional 10T DCIM bit cell at the 3 nm (N3) FinFET technology node.
3) The extended NeuroSim V1.4 framework is used to evaluate macro- and system-level performance metrics of DCIM from 3-nm down to the A5 node.
4) An LUT-based MAC design that eliminates local multipliers and the first stage of the adder tree is introduced at the A5 CFET technology node. It reduces the subarray-level adder tree overhead to achieve dynamic power reduction by 43.7% and footprint by 29.6%, compared to the conventional MAC design (based on the conventional 10T DCIM bit cell) at the A5 CFET technology node.

II. Background

A. Standard Cell Scaling Trend

According to the IRDS device scaling roadmap [8], standard cell area reduction in the recent technology nodes is primarily governed by metal track trimming. However, there exists a limitation in further track reduction due to several physical limitations such as routing congestion and *RC* delay [9]. Consequently, novel device structures such as gate-all-around (GAA) stacked nanosheets have been introduced to maintain scalability [10]. More recently, CFET technology has been considered as a technology breakthrough leveraging vertically stacked nanosheet devices to stack nMOS on top of pMOS [Fig. 1(a)] [11], [12], [13], [14]. Design-technology co-optimization (DTCO) studies have identified that CFET technology will roughly enable the scaling of standard cells and SRAM by 50% with 50% track reduction [13].

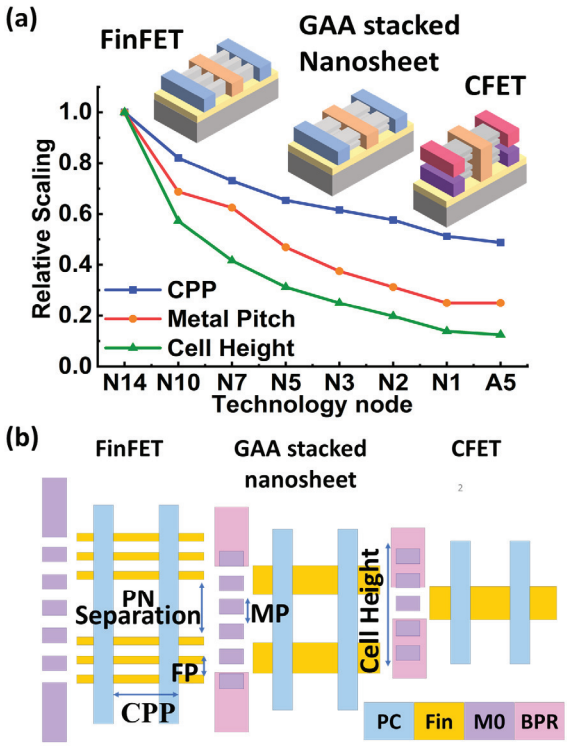


Fig. 1. (a) IRDS scaling roadmap and (b) standard cell layout of FinFET, GAA stacked nanosheet, and CFET.

Fig. 1(a) shows the standard cell scaling trend supported in the extended NeuroSim V1.4 framework introduced in this work. For the calculation of the standard cell area, we assume that a single diffusion break (SDB) between adjacent standard cells and buried power rails (BPRs) is adopted from the 2 nm (N2) node and beyond [9], [10], [15], [16], [17]. As indicated in Fig. 1(a) and (b), vertical stacking can drive further standard cell scaling by enabling aggressively scaled metal tracks [15], [16], [17].

B. DCIM and Its Demonstrations at the Advanced Nodes

DCIM architectures support regular SRAM memory access and a DCIM mode for MAC operations. Fig. 2 presents a schematic view of a typical DCIM architecture. As shown in Fig. 2, the input vector is loaded in parallel and activates multiple rows for parallel multiplication at each bit cell. The adder tree then accumulates the partial sum. DCIM eliminates the need for an analog-to-digital converter (ADC), which bottlenecks ACIM PPA [1], by performing computation entirely in the digital domain, ensuring robust signal integrity and scalability at advanced technology nodes. Moreover, the

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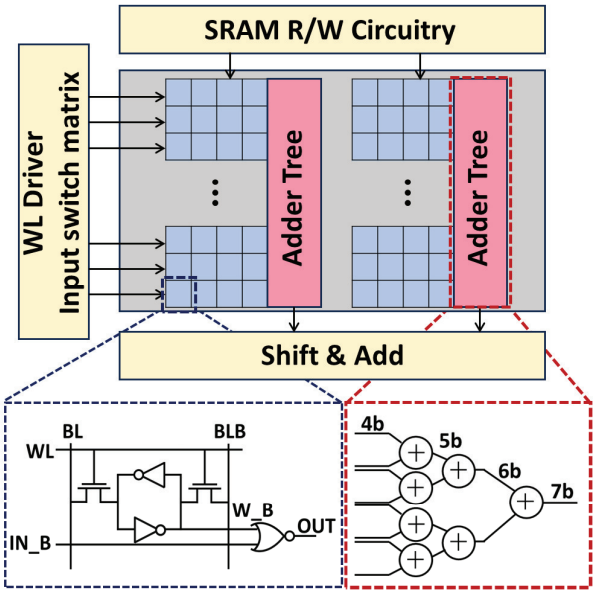


Fig. 2. Schematic view of DCIM array architecture with SRAM + NOR gate bit cell as originally proposed in [4].

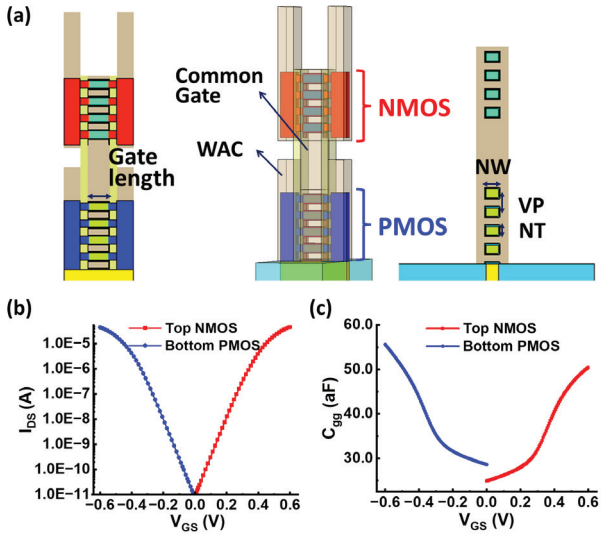


Fig. 3. (a) 3-D CFET device structure with design parameters in Sentaurus TCAD. (b) I_D–V_G. (c) C_{gg}–V_G of top nMOS and bottom pMOS of CFET at V_{DD} = 0.6 V.

parallelism inherent in DCIM architecture allows it to maintain higher throughput without compromising energy efficiency compared to the analog counterpart [7]. In the original design proposed by TSMC [4], a conventional 10T DCIM bit-cell (6T SRAM +4T NOR gate) design is used for 1-b weight storage and bitwise multiplication with 1-b input. The prototype macros employing such DCIM mode (or its variant) have been demonstrated in 22-nm [4], 12-nm [18], [19], 4-nm [20], and 3-nm [5] nodes by companies such as TSMC and MediaTek. By replacing analog MAC operations with digital MAC, DCIM in these studies achieved higher accuracy from high precision and noise immunity, which is critical for modern workloads.

It should be noted that DCIM implementation is compatible with leading-edge foundry processes while taking advantage of digital logic for reliable and precise computation. Thus, DCIM offers a scalable and accurate solution for AI hardware acceleration by capitalizing on the miniaturization and robustness of digital logic. Such features of DCIM make it particularly well suited for handling large-scale DNN workloads, making it an ideal choice for high-performance computing environments, such as data centers [4].

III. Proposed 3-D DCIM Macro With A5 CFET

A. Device Modeling

Utilizing the transistor specifications summarized in Table I based on IRDS 2022 [8], we designed a CFET transistor at the A5 technology node through Sentaurus TCAD simulation [Fig. 3(a)]. The CFET is proposed as a promising device architecture that can meet the demand of a smaller transistor footprint while maintaining a high performance beyond GAA stacked nanosheet transistors [11], [12], [13], [14]. It leverages the complementary nature of CMOS logic by integrating nMOS and pMOS devices controlled by the same gate. In this design, the nMOS device is positioned on top, while the pMOS device is placed on the bottom. This top nMOS and bottom pMOS configuration is chosen to exploit the advantages of strain engineering, particularly at scaled SRAM cell size [13], [21].

TABLE I

Transistor and Standard Cell Design Specifications for A5 CFET. (a) Transistor Specification. (b) Standard Cell Design Specifications

(a)	
NS thickness [NT] (nm)	4
NS width [NW] (nm)	6
NS vertical pitch [VP] (nm)	9
# of stacked NSs	4
Gate length (nm)	12
Equivalent oxide thickness [EOT] (nm)	0.5
Gate height over fin [GHOF] (nm)	3
Spacer width [SW] (nm)	4
Dielectric constant of interlayer dielectric	2.7
(b)	
Metal track in M0 (T)	4.5
Metal pitch (nm)	16
Cell height [CH] (nm)	72
Contacted Poly Pitch [CPP] (nm)	38
Fin pitch [FP] (nm)	23

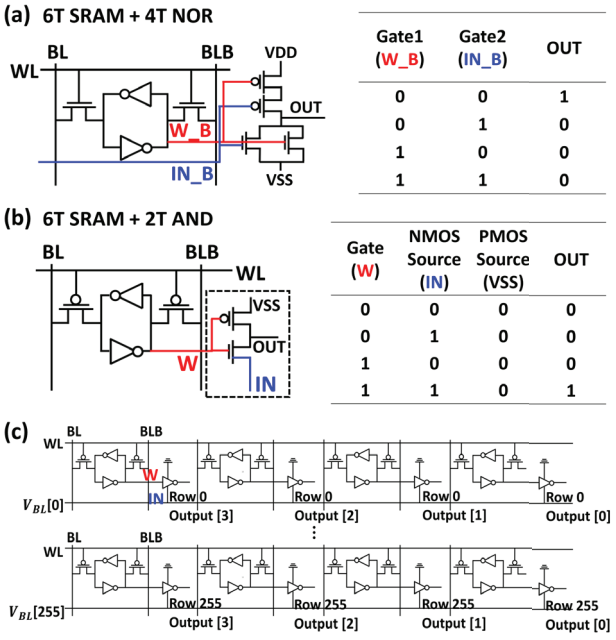
In the TCAD device modeling, to ensure sustained improvements in intrinsic delay and energy consumption with technology scaling, we make selective adjustments to our original device parameter assumptions, significantly reducing the nanosheet vertical pitch (VP) and gate height over fin (GHOF) to reduce middle-of-line (MOL) parasitic capacitance [7]. Based on the designed TCAD device structure illustrated in Fig. 3(a), we conducted I_D–V_G and C_{gg}–V_G simulations to extract the electrical parameters used to implement technology libraries in extension to NeuroSim V1.4 [Fig. 3(b) and (c)].

Our TCAD simulations incorporate models such as the density gradient for quantum confinement, Lombardi and thin layer for mobility degradation, and high-field saturation and

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TABLE II
TRANSISTOR ELECTRICAL PARAMETERS FOR EXTENDED NEUROSIM V1.4

Electrical Parameters	A5 Technology node
V_{DD} [Supply Voltage] (V)	0.6
I_{on} [On current] (μA/μm)	584.4
I_{off} [Off current] (μA/μm)	1.10E-04
g_m (mS/μm)	3.33
C_{gg} [Gate capacitance] (fF/μm)	6.30E-04
Intrinsic delay (= V_{DD} C_{gg}/I_{on}) (ps)	6.46E-01
Intrinsic energy consumption (= $C_{gg}V_{DD}^2$) (fJ/μm)	1.81E-05



Gate1 (W_B)	Gate2 (IN_B)	OUT
0	0	1
0	1	0
1	0	0
1	1	0

Gate (W)	NMOS Source (IN)	PMOS Source (VSS)	OUT
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1

Fig. 4. Schematic design and truth table of (a) conventional SRAM + NOR gate and (b) SRAM with GDI-based 2T AND gate DCIM bit cell using top nMOS and bottom pMOS CFET technology. (c) DCIM macro array with SRAM +2T AND gate DCIM bit cell.

ballistic models to capture electron transport in advanced transistors. In addition, we used Shockley-Read–Hall recombination, slotboom bandgap narrowing, and the deformation potential model to simulate stress effects on the bandgap and density of states. Transport parameters were calibrated to achieve near symmetry in I_D–V_G and C_{gg}–V_G between nMOS and pMOS.

After determining the calibrated process parameters, such as work function, S/D doping concentration, and saturation velocity, to meet the target current, we extracted electrical parameters to extend the library of NeuroSim V1.4 to include the A5 CFET technology node. Transconductance (g_m) is used to estimate latency in circuit components such as SRAM sense amplifiers, while extracted gate and junction capacitance values are utilized to calculate total node capacitance for energy and latency evaluation. The extracted electrical parameters used in NeuroSim V1.4 are summarized in Table II.

B. Schematic Design of the 3-D 8T DCIM Bit Cell

We propose a novel 3-D 8T DCIM bit cell (6T SRAM and 2T AND gate combined together) optimized for the A5 CFET 3-D architecture using the technology parameters detailed in Tables I and II. This design achieves the same functionality as the baseline 10T DCIM bit cell with 6T SRAM and 4T NOR gate [4] while significantly improving area and latency efficiency. Fig. 4(a) and (b) compares the schematic of the proposed 3-D 8T DCIM bit cell with the baseline 10T DCIM bit-cell design. The proposed 8T DCIM bit cell [Fig. 4(b)] leverages A5 CFET technology to enhance area efficiency by reducing device count and interconnect overhead. By utilizing a monolithic CFET inverter configuration, the 3-D 8T DCIM achieves a smaller footprint compared to the 10T DCIM for CFET-based architectures, improving scalability for advanced nodes. The bit cell utilizes gate diffusion input (GDI) logic, which reduces the transistor count compared to standard CMOS designs by integrating transistors and diffusion regions [22]. Although GDI logic reduces output swing, it enables a more compact design and lowers input capacitance.

Another key advantage of the proposed 3-D 8T DCIM design is the enhancement of switching speed and power efficiency by the reduction of input gate capacitance. Using the source (of MOSFET) as the input node lowers capacitance, leading to faster signal propagation and lower dynamic power consumption. In conventional DCIM architecture, the critical delay path includes three main components: bitline (BL) delay, adder tree delay, and the delay of the final NOR gate in a row [23]. The NOR gate input capacitance, combined with metal wire resistance and capacitance, forms a distributed *RC* network along the BL, making it a critical factor in overall latency. Our proposed GDI-based 2T AND gate addresses this challenge by replacing the traditional NOR gate, significantly reducing the logic gate input capacitance. Unlike a typical gate configuration where both the input and weight signals contribute to gate capacitance, the proposed design connects the input to the source of the nMOS and the weight to the gate of the monolithic CFET. This reduces the total input capacitance by introducing a source-side capacitance, which is inherently smaller than gate capacitance. This reduction in input capacitance directly lowers the *RC* delay of the BL and improves latency efficiency.

Furthermore, Fig. 4(c) presents a 256×4 DCIM array using the proposed 8T DCIM bit cell, which is compared against the baseline SRAM + NOR gate DCIM array. By addressing key latency bottlenecks and offering a compact design, the proposed architecture demonstrates its potential for scalable and energy-efficient operation in advanced CFET-compatible technology nodes.

C. Layout Design of the Proposed 3-D 8T DCIM Bit Cell

The 3-D DCIM structure based on the proposed 3-D 8T DCIM bit cell, including the top- and bottom-tier layouts, is shown in Fig. 5. Fig. 5(a) presents the top- and bottom-tier layout designs of the proposed 3-D 8T DCIM bit cell. The layout reflects the innovations in device and interconnect

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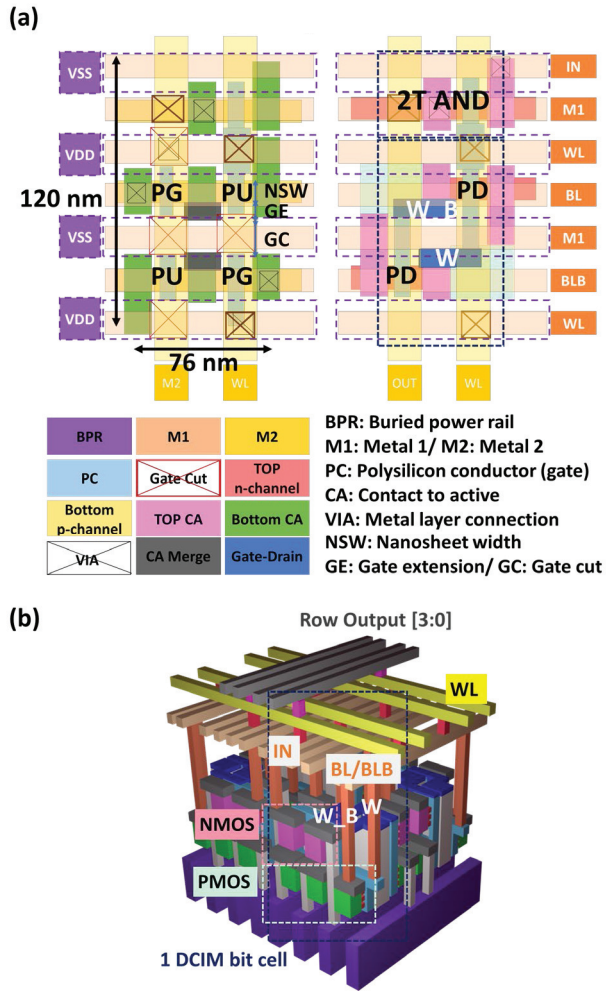


Fig. 5. (a) Bottom tier and top tier of the proposed 3-D DCIM bit-cell (SRAM +2T AND gate) layout. (b) Mini 2 × 2 CFET 3-D DCIM array.

designs enabled by the CFET architecture. Unlike conventional 2-D 6T SRAM designs that utilize nMOS pass gates, the proposed 3-D CFET-based 6T SRAM (a component of 8T DCIM cell) employs pMOS pass gates as illustrated in Fig. 5(a) and (b). In this design, additional process steps for recessing the top nMOS device and leaving the bottom pMOS device isolated enable cross-coupled formation with the gate-to-drain connection, which aligns with the CFET-based 6T SRAM architecture proposed by IMEC [13].

To enhance DCIM cell scalability and mitigate wordline (WL) and BL resistance challenges, we assume the integration of BPR technology [15], [16], [17]. BPR is particularly advantageous in A5 node designs, reducing interconnect resistance and providing enhanced power delivery efficiency, which is critical for high-density memory arrays.

The design rule parameters, including gate extension (GE) and tip-to-tip spacing, are based on state-of-the-art A5 node SRAM designs reported by IMEC [24]. These parameters ensure adherence to scaling constraints while maintaining reliable device operation. While research from IMEC highlights the feasibility of achieving a 3T cell height for 3-D CFET-based 6T SRAM designs through aggressive DTCO [25], we

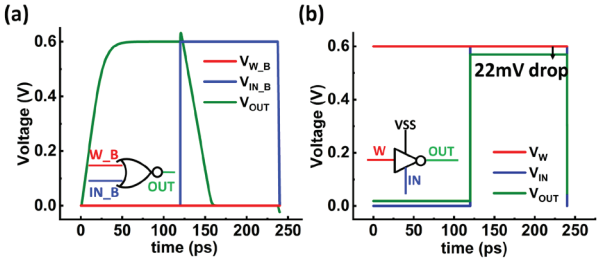


Fig. 6. Function verification with Sentaurus TCAD mixed-mode simulation modeling of (a) SRAM + NOR and (b) SRAM +2T AND. To verify the transition output behavior of the DCIM unit cell, we treat the weight of the SRAM as a static node with V_{DD}.

conservatively assumed a 5T cell height for our SRAM and a 7.5T cell height for the 3-D 8T DCIM bit cell to account for the additional logic overhead and routing complexity. This conservative assumption aligns with industry trends and provides a realistic estimate of the physical design limits for CFET-based 3-D DCIM architectures.

Fig. 5(b) illustrates the 3-D architecture of a mini 2 × 2 CFET 3-D DCIM array, demonstrating the vertical integration and interconnectivity of the top and bottom tiers. The architecture includes four horizontal M3 metal wires allocated for output connections of each row [3:0], which are routed to adder trees. Our proposed 3-D DCIM design harnesses the intrinsic area efficiency of the monolithic CFET by integrating the local multiplier (2T AND) within a single transistor footprint, which cannot be realized through conventional DCIM bit-cell designs or 2-D technology.

D. Validation of the Proposed 3-D 8T DCIM Bit Cell

The proposed 3-D 8T DCIM bit-cell functionality is validated through TCAD mixed-mode simulation, with results compared to the baseline 10T DCIM design [Fig. 4(a) and (b)]. In Fig. 6, while the GDI-based 2T AND gate introduces a modest 22-mV reduction in output swing due to threshold voltage drops, the subsequent adder tree effectively compensates for this.

In traditional DCIM architecture, the input capacitance of the NOR gate significantly contributes to BL loading and interconnect energy consumption, as highlighted in prior work [7], particularly under high toggling activity and at scaled technology nodes. In the proposed GDI-based 2T AND gate, the reduction in input capacitance improves latency by lowering the *RC* delay. Input capacitance reduction also decreases energy consumption, particularly in scenarios involving frequent toggling. By mitigating this critical source of energy overhead, the proposed design demonstrates its potential to enhance the performance and scalability of DCIM architectures in advanced CFET-compatible technology nodes.

IV. PPA Benchmark of DCIM Macro and System

For quantitative assessment of the benefits, we evaluate the macro- and system-level performance metrics of the proposed 3-D DCIM design based on a 3-D 8T DCIM bit cell. Based on the updated A5 CFET technology parameters and designs

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introduced in Section III, we extend the NeuroSim V1.4 framework [7] to incorporate the A5 CFET technology library. PPA analysis of the proposed DCIM array at both macro and system levels is conducted in the extended NeuroSim V1.4. The assumed macro for macro-level simulation is a 256 × 256 DCIM array with integrated peripheral circuits, including adder trees, buffers, and shift-add units. It should be noted that the dynamic energy consumption of the circuits, which is a function of toggling rate, intrinsic device/MOL capacitance (derived from TCAD simulation), and interconnect capacitance (assumed as ~200 aF/μm), is calculated by the estimation methodology introduced in [7]. The interconnect metal is assumed to be copper, which has an established theoretical relationship between the metal dimension and the resistivity. The resistivity is decided by the analytical formula derived in [7]. For system-level evaluation, multiple macros are interconnected to construct a system for full-network inference. VGG-8 with the CIFAR-10 dataset is used as the benchmark for system-level inference workloads, representing a typical lightweight DNN for evaluating memory-intensive and compute-bound operations. It should be noted that all the circuit components are implemented at the A5 CFET technology node following our previous work's simulation assumptions [7], with device metrics from TCAD simulations (Tables I and II) updated to the technology library for PPA evaluations. In our previous work [7], we validated our simulation methodology [78-TOPS/W at 0.84 V, toggling rate (TR) =25%, 29-TOPS/mm^2] by demonstrating agreement with measured silicon data from 22-nm DCIM prototype [4] [89-TOPS/W at 0.84 V, TR =18%, 16.3-TOPS/mm^2], confirming the alignment in hardware PPA estimation trends. Our framework exhibits a consistent trend [618.7-TOPS/W at 0.7 V, TR =6.25%, 381.2-TOPS/mm^2] for the state-of-the-art N3 DCIM prototype [484-TOPS/W at 0.55 V, TR =10%, 493.5-TOPS/mm^2] [5]. This evidences that the extended NeuroSim V1.4 serves as a predictive and reliable benchmarking tool for emerging DCIM technologies.

A. 3-D DCIM Macro-Level Benchmark

Fig. 7 shows a detailed comparison of the macro-level performance results using the extended NeuroSim V1.4 framework, benchmarking across N3 FinFET, 2–1-nm (N2–N1) GAA stacked nanosheet, and A5 CFET technology nodes. Conventional 10T DCIM (6T SRAM + 4T NOR) bit cell [schematic design in Fig. 4(a)] is considered for N1–N3 and A5(a). For A5(b), the proposed 3-D 8T DCIM (6T SRAM +2T AND) bit cell [schematic design in Fig. 4(b)] is considered for the simulation. The analysis evaluates key performance metrics, including area, delay, energy, throughput (TOPS/W), density (TOPS/mm^2), and the figure of merit (FOM) (=TOPS/mm^2 × TOPS/W). A 4 × 4 b MAC operation is defined as a single operation for consistency across comparisons.

The critical path in the macro is dominated by the adder tree. The improvements in the A5 CFET technology node's delay metrics (Table II) contribute to the enhanced macro speed with the proposed 3-D DCIM architecture at the A5 node. Moreover, the proposed 3-D 8T DCIM bit cell reduces

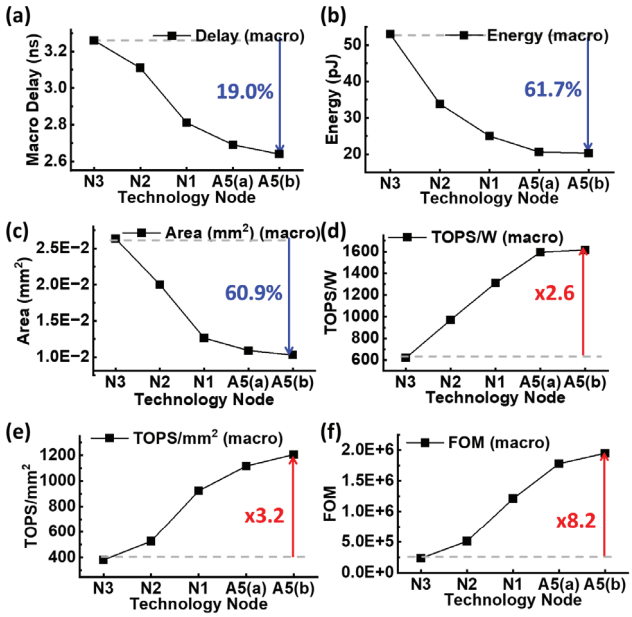


Fig. 7. Macro-level performance analysis of DCIM. (a) Critical delay. (b) Energy. (c) Area. (d) TOPS/W. (e) TOPS/mm^2. (f) FOM = TOPS/W × TOPS/mm^2.

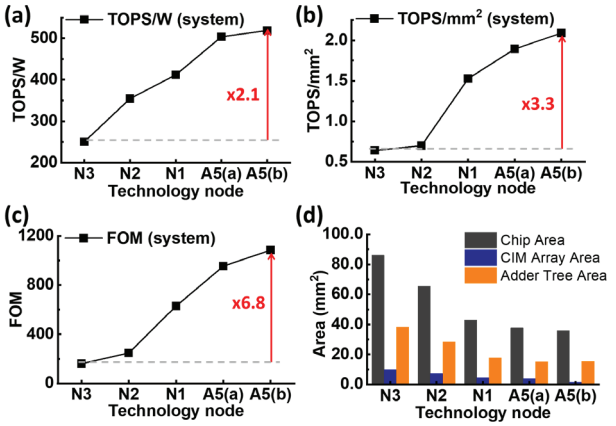
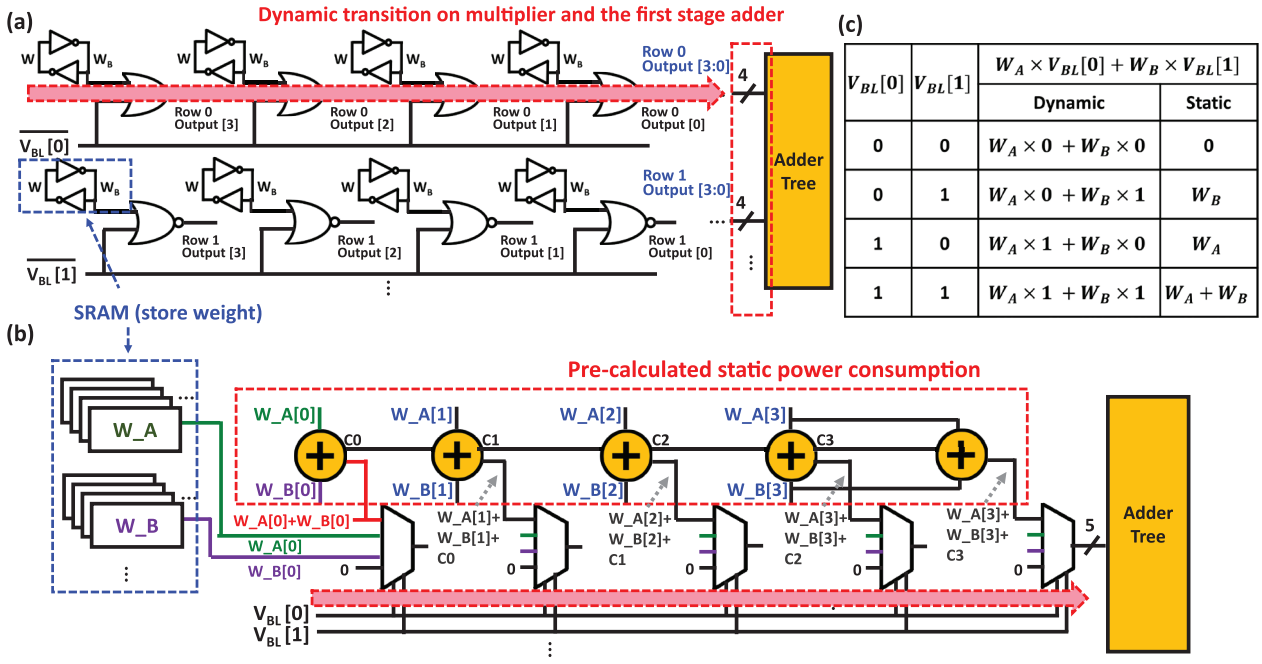


Fig. 8. System-level performance analysis of DCIM. (a) TOPS/W. (b) TOPS/mm^2. (c) FOM = TOPS/W × TOPS/mm^2. (d) Area comparison of chip, CIM array, and adder tree.

input gate capacitance through its compact GDI-based design, lowering BL loading and overall macro latency compared to the baseline design with NOR gate. It also minimizes switching energy, addressing key contributors to macro energy consumption, including adder trees and interconnects. Due to the scaled device area and 3-D stackable nature of the A5 CFET technology, the proposed DCIM bit cell achieves smaller standard cell and macro areas, leading to higher macro density and improved energy efficiency [Fig. 7(a)–(c)].

Macro-level performance evaluations [Fig. 7(d)–(f)] demonstrate significant TOPS/W, TOPS/mm^2, and FOM improvements. The proposed 8T DCIM bit-cell design [A5(b)] at the A5 CFET technology node achieves 8.2× improvements in FOM over the conventional 10T DCIM bit-cell design at the N3 FinFET technology node. A5(b) achieves 1.35%, 8.10%,

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$V_{BL}[0]$	$V_{BL}[1]$	$W_A \times V_{BL}[0] + W_B \times V_{BL}[1]$	
		Dynamic	Static
0	0	$W_A \times 0 + W_B \times 0$	0
0	1	$W_A \times 0 + W_B \times 1$	W_B
1	0	$W_A \times 1 + W_B \times 0$	W_A
1	1	$W_A \times 1 + W_B \times 1$	$W_A + W_B$

Fig. 9. (a) Conventional 10T DCIM bit-cell design with dynamic transitions at NOR gate multiplier and the first stage of the adder tree. (b) LUT-based DCIM design without dynamic transitions, which consumes only static power at precalculating adder. (c) Static precalculated MAC operation table.

and 9.53% better performance of TOPS/W, TOPS/mm^2, and FOM, respectively, over the conventional 10T DCIM bit-cell design at A5 CFET technology node [A5(a)]. We further evaluated ResNet18 and ResNet24 on the ImageNet dataset for a macro-level benchmark. A5(b) CFET-based DCIM achieves up to 9× improvement in FOM compared to N3 FinFET-based DCIM design (data not included in this article). These results highlight the scalability and energy efficiency of the proposed 3-D DCIM architecture at advanced technology nodes (below 1 nm).

B. 3-D DCIM System-Level Benchmark

Fig. 8 compares system-level metrics across different nodes, showing significant improvements with scaling to the A5 CFET technology node. Conventional 10T DCIM (6T SRAM +4T NOR gate) bit cell [schematic design in Fig. 4(a)] is considered for N1–N3 and A5(a). For A5(b), the proposed 3-D 8T DCIM (6T SRAM +2T AND gate) bit cell [schematic design in Fig. 4(b)] is considered for the simulation. The proposed 3-D 8T DCIM bit-cell design [A5(b)] achieves 6.8× improvements in FOM over the conventional 10T DCIM bit-cell design at the N3 FinFET technology node, primarily from reduced macro and interconnect energy [Fig. 8(a)–(c)]. Our proposed 3-D 8T DCIM bit-cell design [A5(b)] achieves 2.97%, 10.43%, and 13.70% better performance of TOPS/W, TOPS/mm^2, and FOM, respectively, over the conventional 10T DCIM bit-cell design [A5(a)] at A5 CFET technology node at the system level. Despite a ~50% reduction in CIM array area, chip area reduction is modest (~20%) due to the adder tree's dominant area contribution (>43%) [Fig. 8(d)]. This indicates that developing more area-efficient adder tree designs is essential for further reducing chip area, which could improve energy and speed by shortening the total chip wiring length.

V. LUT-Based DCIM Design

A. LUT-Based DCIM Design

As previously discussed, the adder tree presents a significant bottleneck to macro- and system-level areas and dynamic power consumption. To overcome this bottleneck, we present an LUT-based MAC DCIM design [18] implemented at the A5 CFET technology node as a replacement for a local multiplier and the first stage of the adder tree [Fig. 9(a) and (b)], aiming to further reduce chip area and dynamic power consumption. For simulation, we adopt the same circuit design as the prior work [18], but the underlying circuit components are simulated with A5 CFET technology node parameters obtained from Section III. According to Fig. 9(c), when input BL [1:0] is 11, the adder output will be (W_A + W_B), which is preadded in advance. By precalculating the MAC operation (W_A × V_BL [0] + W_B × V_BL [1]) and transferring data to the digital MUX, dynamic transitions during input changes are eliminated, effectively mitigating the high power consumption associated with traditional multiplier and adder circuits. This is feasible thanks to DCIM's weight stationary feature, as only input is used to look for the MAC values precalculated before the adder tree's next stage.

It should be noted that the LUT-based DCIM removes local multipliers typically implemented with 4T NOR gates (or 2T AND gates in this work), eliminating their dynamic power consumption. Unlike conventional 10T or the 3-D 8T DCIM bit-cell design proposed earlier, which is focused on reducing PPA overhead from bitwise multiplication, the LUT-based MAC approach reduces the PPA overhead from both bitwise multiplication and subarray-level MAC operation. The energy consumption of the proposed LUT-based DCIM design is modeled based on toggling rate and energy parameters

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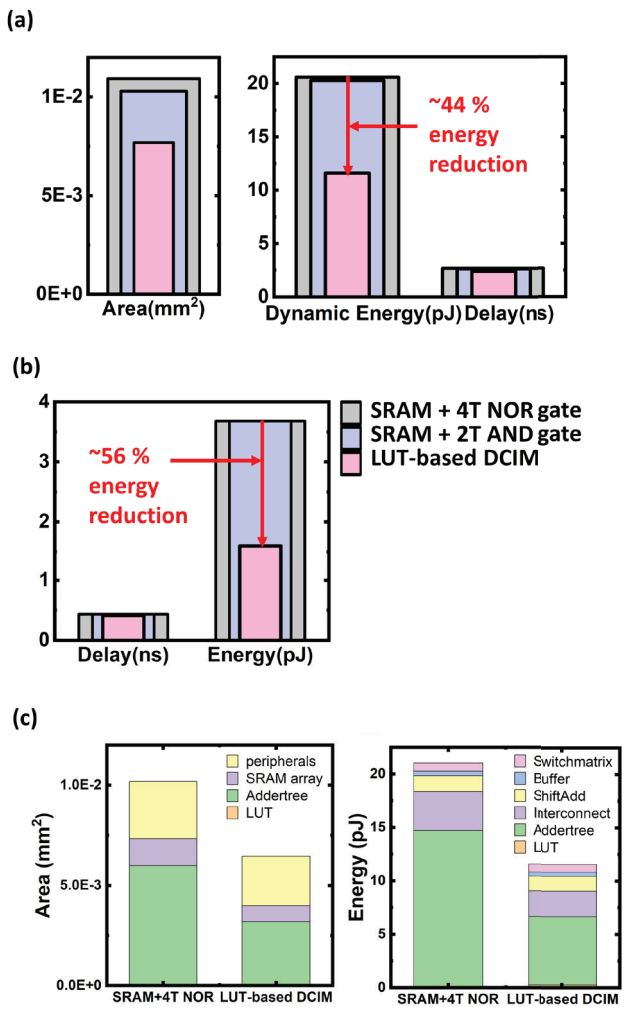


Fig. 10. Macro-level performance analysis of conventional 10T DCIM (SRAM +4T NOR) [Fig. 4(a)], proposed 8T DCIM (SRAM +2T AND) [Fig. 4(b)] [Fig. 9(a)] and LUT-based DCIM [Fig. 9(b)] at A5 CFET technology node. (a) Macro area, dynamic energy, and critical delay. (b) Adder tree delay and dynamic energy. (c) Macro area and energy breakdown to clarify the optimization sources.

following the NeuroSim V1.4 framework [7]. This approach shrinks the total chip area and enhances energy and speed performance, thereby fully leveraging the benefits of A5 CFET and DCIM technology.

B. Macro-Level Analysis With LUT-Based DCIM Design

To demonstrate the benefits, we conduct macro-level simulations using extended NeuroSim V1.4, demonstrating that the LUT-based DCIM design achieves significant reductions in dynamic power consumption while also reducing the total macro area. Simulation results demonstrate that the MUX-based approach to LUT-based DCIM design offers substantial improvements over traditional DCIM adder tree and multiplier-based designs [Fig. 10(a) and (b)].

At the A5 CFET technology node, the LUT-based MAC operation design achieves a 29.6% macro area reduction and 11.2% macro delay improvement compared to the conventional MAC operation design with the 10T DCIM bit cell. This area reduction stems from the compact MUX-based architecture replacing the multiplier and the first stage of the adder tree, which minimizes the transistor count required for lookup-table functionality. In terms of dynamic energy consumption, it achieves a significant reduction of 43.7% compared to the conventional MAC design with the 10T DCIM bit cell, which is primarily attributed to the elimination of dynamic transitions in MAC operations [Fig. 10(c)]. By precalculating results and storing them in the LUT, this approach significantly decreases switching activity, a primary contributor to dynamic power [Fig. 10(a)]. This can be explained, as shown in Fig. 10(b), which shows over 50% improvement in energy consumption in adder tree operations, typically one of the most energy-intensive components in DCIM architectures.

The presented results highlight the effectiveness of the LUT-based MAC in optimizing area, delay, and power consumption metrics. It not only improves the PPA metrics from 3-D DCIMs based on the conventional 10T DCIM bit cell and the 3-D 8T DCIM bit cell (proposed earlier) but also enhances the scalability of CFET-based architectures by reducing the number of transistors required for MAC operation. These findings underscore the potential of the proposed 3-D DCIM architectures as a promising direction for next-generation DCIM systems at A5 CFET technology, particularly for AI workloads where energy efficiency and compactness are critical. While the logic-level concept of LUT-based DCIM design is node-agnostic and has been previously demonstrated in advanced nodes such as N3 [5], our work explores and highlights its scalable integration within CFET-based 3-D architectures utilizing the device-to-system codesign framework (from physics-level simulation to system-level simulation). By leveraging the compact SRAM enabled by CFET device architecture by following IRDS and IMEC's A5 design rules, we demonstrate enhanced PPA benefits and improved architectural scalability. The demonstrated results not only indicate the reduced number of transistors required for MAC operation but also highlight the practical feasibility of LUT-based MACs in future CFET technology nodes.

VI. Potential Challenges of DCIM Design With A5 CFET Technology

While the simulated macro- and system-level PPA results show great promise for the proposed DCIM architectures, there are several anticipated challenges in the practical implementation. Thus, we highlight some critical challenges that must be importantly considered for future A5 CFET-based DCIM technology development. The feasibility of the proposed DCIM architectures will be dependent on the viability of the A5 CFET process. CFET process integration is reliant on advanced process steps, such as dielectric selective deposition on metal and high aspect ratio trench etch [26], [27]. Increased process complexity posed by advanced process requirements at a highly scaled node (A5) might result in larger susceptibility to process variation, potentially limiting system-level performance. Moreover, pronounced self-heating in CFET nanosheet devices could be detrimental to the current drivability of the transistors in the DCIM cell and adder trees, degrading the system-level latency performance [28], [29].

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Due to the close proximity of the 2T AND to 6T SRAM in the proposed 3-D 8T DCIM bit cell, intradevice thermal crosstalk between 6T SRAM and 2T AND could also affect the stability of the SRAM. Given that stacked nanosheet exhibits reliability concerns, including hot carrier injection (HCI) and bias temperature instability (BTI), CFET is also expected to face similar challenges [30]. These reliability issues necessitate reliability-aware design considerations to mitigate long-term performance degradation [31]. To ensure the successful development of the proposed 3-D DCIMs, future process advancements must address these challenges.

VII. Conclusion

This article presents a device-to-system codesign (from physics-level simulation to system-level simulation) process for developing two novel 3-D DCIM architectures enabled by the adoption of the A5 CFET technology node. The proposed 3-D DCIM architectures are based on: 1) conventional MAC with 3-D 8T DCIM bit cell (6T SRAM +2T AND) and 2) LUT-based MAC replacing local multipliers and the first stage of the adder tree. The A5 CFET device model has been simulated and calibrated through Sentaurus TCAD simulation. The calibrated A5 CFET device has been used to implement a novel 3-D 8T DCIM bit cell and peripheral circuit for the proposed 3-D DCIM architectures. For quantitative evaluation of the proposed 3-D DCIM architectures, an extended NeuroSim V1.4 framework has been developed, which allows the macro- and system-level PPA evaluation of CIM systems implemented with A5 CFET technology. It has been shown that the proposed 3-D DCIM architectures with 3-D 8T DCIM bit cell or LUT-based MAC operation significantly improve the macro- and system-level PPA metrics from the prior designs. These results underscore the potential of CFET technology to address scalability challenges at advanced nodes and enhance DCIM performance. The proposed 3-D DCIM architectures and the extended NeuroSim V1.4 framework establish a strong foundation for developing high-performance AI accelerators with CFET technology and suggest significant implications for technology scaling below 1 nm for next-generation DCIM design.

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